Lecture Notes on Thermodynamic Computing

Chapter 3: Stochastic Computing Architectures

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1 3.1 Introduction to Stochastic Computing

• Stochastic Number Representation

Definition 1.1 (Stochastic Number)**.** A stochastic number X is represented by a random bit stream where the probability of bit '1' equals the value:

$$P(b_i = 1) = X, \quad X \in [0, 1] \tag{1}$$

Key properties:

- Natural fault tolerance
- Simplified arithmetic operations
- Inherent noise immunity
- Compact hardware implementation

• Stochastic vs Binary Arithmetic

Operation	Binary	Stochastic
Addition	Multi-bit adder	Scaled OR gate
Multiplication	Multiplier array	Single AND gate
Square root	Complex algorithm	JK flip-flop
Sigmoid	Lookup table	Comparator

Table 1: Arithmetic complexity comparison

• Thermal Noise as Random Source

Johnson noise in resistors provides natural randomness:

$$\langle V_n^2 \rangle = 4k_B T R \Delta f \tag{2}$$

For bit-stream generation at frequency f:

$$SNR = \frac{V_{signal}^2}{4k_B T R f} \tag{3}$$

Required signal amplitude:

$$V_{signal} > 4\sqrt{k_B T R f} \tag{4}$$

2 3.2 Stochastic Number Generators

• Thermal Noise-Based Generators

Circuit implementation using subthreshold MOSFETs:

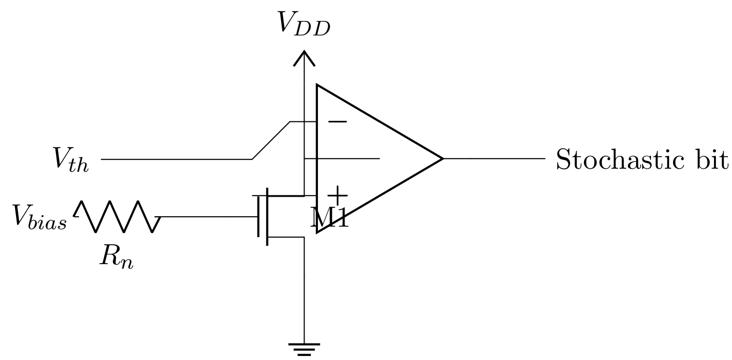

Figure 1: Thermal noise stochastic generator

Current fluctuations in subthreshold:

$$\sigma_I^2 = 2qI_{dc}\Delta f \tag{5}$$

Bit generation probability:

$$P(1) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{I_{bias} - I_{th}}{\sigma_I\sqrt{2}}\right)\right] \tag{6}$$

• Linear Feedback Shift Registers (LFSR)

Deterministic pseudo-random generation:

$$x_{n+1} = a_1x_n \oplus a_2x_{n-1} \oplus ... \oplus a_kx_{n-k+1} \tag{7}$$

Maximum sequence length: $2^k - 1$ bits
Energy per bit:

$$E_{LFSR} = k \cdot C_{ff}V_{DD}^2 \tag{8}$$

where C_{ff} is flip-flop capacitance.

• Analog-to-Stochastic Converters

Convert analog voltage to stochastic stream:

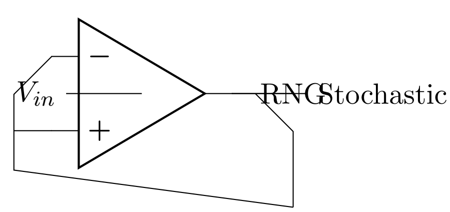

Figure 2: Analog-to-stochastic converter

Conversion accuracy:

$$\epsilon = \frac{\sigma}{\sqrt{N}} = \sqrt{\frac{X(1-X)}{N}} \tag{9}$$

where N is bit stream length.

3 3.3 Stochastic Arithmetic Units

• Multiplication

Single AND gate implements multiplication:

$$P(Z=1) = P(X=1) \cdot P(Y=1) = XY \tag{10}$$

Hardware cost: 1 gate vs n^2 gates for binary
Energy comparison:

$$\frac{E_{stochastic}}{E_{binary}} = \frac{1}{n^2} \cdot \frac{N}{1} \tag{11}$$

For $N = 256$ bits, $n = 8$: ratio = 4

• Scaled Addition

OR gate with scaling for addition:

$$P(Z=1) = P(X=1) + P(Y=1) - P(X=1)P(Y=1) \tag{12}$$

For small probabilities ($X, Y \ll 1$):

$$Z \approx X + Y \tag{13}$$

Exact addition requires MUX-based circuits:

$$Z = \frac{X+Y}{2} \tag{14}$$

• Square Root

Implemented with JK flip-flop:

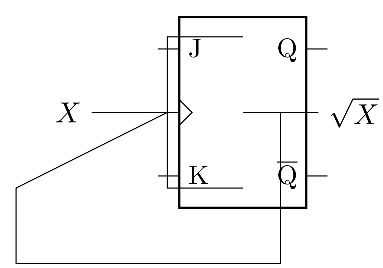

Figure 3: Stochastic square root circuit

Steady-state analysis:

$$P(Q=1) = \sqrt{X} \tag{15}$$

Convergence time:

$$\tau = \frac{1}{X(1-\sqrt{X})} \tag{16}$$

4 3.4 Complex Function Implementation

- **Polynomial Approximation**

Any function $f(x)$ can be approximated as:

$$f(x) = \sum_{i=0}^{n} a_i x^i \tag{17}$$

Stochastic implementation using Bernstein polynomials:

$$f(x) = \sum_{i=0}^{n} f\left(\frac{i}{n}\right) \binom{n}{i} x^i (1-x)^{n-i} \tag{18}$$

- **Sigmoid Function**

Approximate sigmoid using polynomial:

$$\sigma(x) \approx 0.5 + 0.25x - 0.0625x^3 \tag{19}$$

Stochastic circuit uses multipliers and adders in parallel.
Hardware cost: $O(n)$ vs $O(2^n)$ for lookup table.

- **Exponential and Logarithm**

Exponential via iterative squaring:

$$e^x = \lim_{n\to\infty} \left(1 + \frac{x}{2^n}\right)^{2^n} \tag{20}$$

Logarithm via Newton-Raphson:

$$\ln(x) = 2\tanh^{-1}\left(\frac{x-1}{x+1}\right) \tag{21}$$

5 3.5 Error Analysis and Convergence

- **Variance Analysis**

For bit stream of length N:

$$\mathrm{Var}(\hat{X}) = \frac{X(1-X)}{N} \tag{22}$$

Standard deviation:

$$\sigma_X = \sqrt{\frac{X(1-X)}{N}} \tag{23}$$

Worst case at $X = 0.5$: $\sigma_{max} = \frac{1}{2\sqrt{N}}$

- **Correlation Effects**

For correlated inputs with correlation ρ:

$$\mathrm{Var}(XY) = \frac{X(1-X)Y(1-Y)}{N}(1+\rho) \tag{24}$$

Correlation increases error; decorrelation circuits required.

- **Progressive Precision**

Precision improves with bit stream length:

$$\epsilon(N) = \frac{k}{\sqrt{N}} \tag{25}$$

For n-bit precision: $N = 4^n$ bits required
Trade-off: precision vs latency/energy

6 3.6 Thermodynamic Stochastic Processors

- **Subthreshold Stochastic Units**

Operating voltage optimization:

$$V_{opt} = nV_T \ln\left(\frac{2\sqrt{N}}{P_e}\right) \tag{26}$$

where P_e is target error probability.
Energy per operation:

$$E_{op} = \frac{1}{2}CV_{opt}^2 \cdot N \tag{27}$$

- **Parallel Processing Architecture**

Multiple stochastic units in parallel:

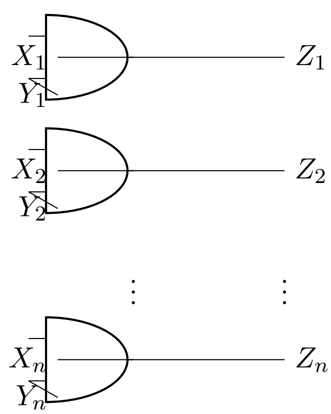

Figure 4: Parallel stochastic multipliers

Throughput scaling:

$$\text{Throughput} = \frac{n \cdot f_{clk}}{N} \tag{28}$$

Power efficiency:

$$\frac{\text{Ops/s}}{\text{Watt}} = \frac{n \cdot f_{clk}}{N \cdot n \cdot P_{unit}} = \frac{f_{clk}}{N \cdot P_{unit}} \tag{29}$$

- **Fault Tolerance**

Bit flip probability in noisy environment:

$$P_{flip} = \frac{1}{2} \exp\left(-\frac{(V_{DD}/2)^2}{2\sigma_{noise}^2}\right) \tag{30}$$

Effect on stochastic number:

$$X_{observed} = X_{true}(1 - 2P_{flip}) + P_{flip} \tag{31}$$

Graceful degradation for small P_{flip}.

7 3.7 Design Optimization

- **Bit Stream Length Optimization**

Trade-off between accuracy and latency:

$$\text{Cost} = \alpha \cdot \frac{1}{\sqrt{N}} + \beta \cdot N \tag{32}$$

Optimal length:

$$N_{opt} = \left(\frac{\alpha}{2\beta}\right)^{2/3} \tag{33}$$

- **Supply Voltage Scaling**

Energy-delay product for stochastic unit:

$$EDP = \frac{1}{2} C V_{DD}^2 \cdot \frac{N}{f_{clk}} \tag{34}$$

Clock frequency in subthreshold:

$$f_{clk} = \frac{I_{ds}}{C V_{DD}} \propto \exp\left(\frac{V_{DD}}{n V_T}\right) \tag{35}$$

Optimal voltage:

$$V_{DD}^{opt} = n V_T \ln\left(\frac{2N}{C V_{DD}^{opt}}\right) \tag{36}$$

- **Temperature Optimization**

Thermal noise scaling:

$$\sigma_{thermal} \propto \sqrt{T} \tag{37}$$

Signal margin:

$$\text{Margin} = V_{signal} - 4\sigma_{thermal} \tag{38}$$

Optimal operating temperature balances noise and leakage.

8 3.8 Applications in Neural Networks

- **Stochastic Neurons**

Activation function implementation:

$$y = \sigma\left(\sum_i w_i x_i + b\right) \tag{39}$$

Stochastic multiplication: $w_i \cdot x_i$ via AND gates
Accumulation: Scaled OR gates or counters

- **Learning Algorithms**

Stochastic gradient descent:

$$w_{new} = w_{old} - \eta \nabla E \tag{40}$$

Gradient estimation using stochastic streams:

$$\frac{\partial E}{\partial w} \approx \frac{E(w + \Delta w) - E(w - \Delta w)}{2\Delta w} \tag{41}$$

- **Dropout Implementation**

Natural dropout via stochastic masking:

$$y = x \cdot \text{Bernoulli}(p) \tag{42}$$

Hardware: Single AND gate with random bit stream.

9 Examples

Example 9.1 (Stochastic Multiplier Energy)**.** Calculate energy consumption for 8-bit multiplication using stochastic vs binary.

Given:

- 8-bit operands
- Stochastic stream length: $N = 256$
- Gate capacitance: $C = 1$ fF
- Supply voltage: $V_{DD} = 0.5$ V

Solution:
Binary multiplier energy:

$$E_{binary} = 64 \times \frac{1}{2} \times 10^{-15} \times (0.5)^2 \tag{43}$$
$$= 8 \times 10^{-15} \text{ J} = 8 \text{ fJ} \tag{44}$$

Stochastic multiplier energy:

$$E_{stochastic} = 256 \times \frac{1}{2} \times 10^{-15} \times (0.5)^2 \tag{45}$$
$$= 32 \times 10^{-15} \text{ J} = 32 \text{ fJ} \tag{46}$$

Energy ratio: $\frac{32}{8} = 4\times$ higher for stochastic
However, stochastic provides inherent fault tolerance.

Example 9.2 (Convergence Analysis)**.** Find required bit stream length for 1
Given:

- Input values: $X = 0.6$, $Y = 0.4$
- Target accuracy: $\epsilon = 0.01$
- Confidence level: 95

Solution:
Expected output: $Z = XY = 0.24$
Variance:

$$\text{Var}(Z) = \frac{Z(1-Z)}{N} = \frac{0.24 \times 0.76}{N} \tag{47}$$
$$= \frac{0.1824}{N} \tag{48}$$

For 95

$$1.96\sqrt{\frac{0.1824}{N}} < 0.01 \tag{49}$$
$$\sqrt{N} > \frac{1.96 \times \sqrt{0.1824}}{0.01} \tag{50}$$
$$N > 3502 \tag{51}$$

Required bit stream length: $N = 3502$ bits

10 Homework Problems

Assignment 3.1: Stochastic Computing Architectures

1. **Number Representation**:
    - Convert 0.375 to stochastic bit stream (length 16)
    - Calculate expected variance
    - Design thermal noise generator circuit
2. **Arithmetic Operations**:
    - Design stochastic divider using AND/OR gates
    - Implement polynomial $f(x) = x^2 + 0.5x$
    - Calculate hardware complexity vs binary

3. **Error Analysis**:
   - Find bit stream length for 0.1
   - Analyze correlation effects in multiplication
   - Design decorrelation circuit
4. **Function Approximation**:
   - Implement $\tanh(x)$ using Bernstein polynomials
   - Compare energy vs lookup table
   - Optimize degree vs accuracy trade-off
5. **Neural Network Implementation**:
   - Design 4-input stochastic neuron
   - Implement ReLU activation function
   - Calculate throughput and power
6. **Subthreshold Design**:
   - Optimize supply voltage for minimum EDP
   - Consider process variations ($\sigma_{V_{th}} = 50$ mV)
   - Design bias circuits for temperature stability
7. **Fault Tolerance Analysis**:
   - Model bit flip probability vs voltage scaling
   - Calculate output error for 1
   - Compare with binary arithmetic resilience
8. **System Integration**:
   - Design mixed-signal interface circuits
   - Implement error correction coding
   - Optimize throughput vs power trade-off

Next Chapter Preview: Chapter 4 will explore "Probabilistic Logic Circuits" including Bayesian networks and uncertainty propagation in hardware.